

SWAPNIL KHARCHE

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EDUCATION

STATE UNIVERSITY OF NEW YORK AT BINGHAMTON

Binghamton, NY

Master of Science in Computer Science

GPA: 3.68/4.00

05/2019

Relevant Courses: Algorithms & Data Structures, Computer Architecture, Operating Systems, Design Patterns, Data Mining.

COLLEGE OF ENGINEERING, PUNE

Pune, MH, India

Bachelor of Science in Electronics and Telecommunications

Relevant Courses: Electronic Devices and Circuits, *Microprocessors and Microcontrollers, Integrated circuits, Digital Signal Processing.*

05/2015

TECHNICAL SKILLS / PROJECTS

- **Programming Languages:** Java (Advanced), C++/C (Intermediate), Python (Beginner), JavaScript (Beginner)
- **Web Technologies:** HTML5 / CSS3 (Beginner), React (Beginner), Spring (Intermediate), REST APIs (Intermediate)
- **Databases:** MySQL (Intermediate), Redis (Beginner), DB2 (Prior Experience)
- **Other Tools/Applications:** Linux, Git, Agile, Docker, AWS, ROS, CI/CD tools, ITIL, SDLC, Printed Circuit Board design, Matlab, VLSI design, Networking protocols, UART / USB/ RS232 protocols, TCP/ IP, Ethernet.
- **CPU Pipeline Simulator Project:** Implemented an assembly instruction pipeline simulator to mimic the behavior of a speculative modern CPU using queues and buffers with different memory implementations such as Cache and In-memory storage. *Utilized – Java, IntelliJ, Git.*
- **Recommendation System Project:** Designed a recommendation system using collaborative filtering on user preference rating. Designed an API with two endpoints to get data on the users and the most preferred item for that user. *Utilized – Python, pandas, Postman.*

EXPERIENCE

APPLIED MATERIALS

Santa Clara, CA

Software Engineer 2 Senior

02/2022 - Present

- **Engineered** wafer scheduling and sequencing algorithms in C++ for physical vapor deposition tools, improving throughput and reducing system downtime by 15%.
- **Collaborated** with global hardware and controls teams to convert customer specifications into robust software modules
- *Technologies: C++, C#, Git, Visual Studio.*

TATA CONSULTANCY SERVICES - STRYKER CORPORATION

Kalamazoo, MI

Software Engineer

10/2019 - 01/2022

- **Teamwork:** Coordinated with multiple cross functional teams for integration and configuration of a cloud based procurement application used by about 4000 company employees, with multiple downstream applications. Developed REST APIs to handle real time requests to get account and purchasing data with MySQL and Redis.
- **End User Focused:** Observed large amounts of manual efforts by the procurement team. Identified root cause of efforts by meeting with business users. Introduced automated triggers based on business requirements to execute data integration to resolve inefficient data gaps, reducing manual efforts by 20%.
- *Technologies: Spring Boot, RESTful APIs, XML, JSON, MySQL, MongoDB, Microservices, GraphQL, Java8.*

VEHWARE

Binghamton, NY

Computer Vision Intern

09/2018 - 12/2018

- **Designed and implemented** a hazard detection model using optical flow and YOLO for real-time traffic video analysis, reducing frame processing time by 30%.
- **Developed** a web API to interface with the model for demo deployment using web2py (MVC architecture).
- *Technologies: OpenCV, C++, Python, TensorFlow, YOLO, MVC.*

ACCENTURE SOLUTIONS

Pune, MH, India

Application Development Analyst

10/2015 - 08/2017

- **Development:** Developed web applications for client server environments and added functionalities to existing applications using Java/J2EE technologies.
- **Data Analysis:** Performed complex SQL queries to generate data reports which lead to correction of multiple issues in the client application.
- *Technologies – Java/J2EE, MySQL, JDBC, Maven.*